

A Game-based Augmented Reality Navigation System to Support Makerspace User Education in a University Library

Ching-Yu Hsieh *

Graduate Institute of Library, Information
and Archival Studies
National Chengchi University
Taipei City, Taiwan, R.O.C.
106155503@nccu.edu.tw

Chih-Ming Chen

Graduate Institute of Library, Information
and Archival Studies
National Chengchi University
Taipei City, Taiwan, R.O.C.
chencm@nccu.edu.tw

Ya-Chu Yang

Graduate Institute of Library, Information
and Archival Studies
National Chengchi University
Taipei City, Taiwan, R.O.C.
enya0818@gmail.com

Abstract—In recent years, a makerspace has been regarded as an important learning field for cultivating students' creative and thinking abilities. However, the diversified operations and applications of makerspace equipment may cause the librarians who are responsible for makerspace user education cannot burden their working load due to the limitation of manpower. Therefore, this study develops a game-based augmented reality navigation system (GARNS) to support makerspace user education by combining a game-based learning method and augmented reality (AR) tools, hoping to promote learners' learning motivation and immersive experience, and enhances the learning performance of makerspace user education. Furthermore, this study also gathered learners' learning behaviors for behavior analysis based on lag sequential analysis to understand the effective learning mode of learners using the GARNS. With a true experimental research method, a total of 24 Grade 11 students from a high school in Keelung City, Taiwan were recruited to participate in the makerspace user education. Among them, a total of 10 students were randomly assigned to the experimental group using the GARNS, the other 7 students were randomly assigned to control group 1 using the web navigation system, and the remaining 7 students were assigned to the control group 2 using the narrative guided tour with librarian to conduct makerspace user education. Analytical results show that learners can achieve significant learning effectiveness no matter using the GARNS, web navigation system, or traditional narrative guided tour with librarian for makerspace user education. There was no significant difference between the GARNS group and the narrative guided tour with librarian group, and between the web navigation system group and the narrative guided tour with librarian group. However, there was a significant difference between the GARNS group and the web navigation system group, and the GARNS group was significantly better than the web navigation system group.

Keywords—Makerspace, Makerspace user education, Augmented reality, Game-based learning, Learning effectiveness

I. INTRODUCTION

As early as 1873, public libraries hosted social gatherings on the theme of sewing or crafting, and in the 20th century, libraries formally set up dedicated spaces for patrons to rent tools or do creative crafting[1]. A makerspace is defined as a learning space for creative activities, providing resources such as crafts, new technologies, tools, or other resources that support users' needs for creativity[2]. With the evolution and transformation of library functions, [3]proposed the concept of library 4.0, defining the concept of creative space as "infinite creative space,"

the key point of which is to enable students to not only extend the knowledge learned in a class but also develop the ability to create new knowledge[4]. In recent years, providing hands-on creative services in the makerspace in a library has been considered an innovative patron service[5].

A makerspace is regarded as a place to enrich design thinking[6] and a place to expose students to implement and practice the application of knowledge related to science, technology, engineering, and mathematics (STEM)[7]. To enable users to make full use of a makerspace in a library for creativity, it is necessary to develop education on the use of a library's makerspace [8]. However, due to the diverse equipment and applications in a library's makerspace and the different learning needs of individual users, librarians often feel overwhelmed when conducting education on the use of a library's makerspace. Therefore, if we can develop digital learning methods in a library's makerspace and allow users to use mobile devices to learn by themselves, it will be possible to not only effectively save library teaching manpower but also enhance users' willingness to use a library's makerspace.

Traditional library user education is usually conducted through librarian-guided tours and instruction, but with the development of information and communication technology (ICT), the traditional library user education mode is gradually toward technology-based learning mode, such as the use of web-based technology, augmented reality (AR), and tangible user interface (TUI). For example, [9] used interactive augmented reality to support library user education in an elementary school and showed that this learning model improved learning satisfaction and motivation. Therefore, in developing the learning mode of makerspace user education, besides using traditional lecture mode, using the advantages of interactive technology to develop the autonomous learning mode can not only save the manpower burden but also has the potential to enhance the learning effectiveness and motivation of learners.

In addition to the use of interactive technology to support the use of a makerspace, the game-based learning model has been widely used in library user education [10] and various teaching fields [11], and has been confirmed to be effective in promoting learning motivation, sustained attention, and interest [12]. Compared to traditional teacher-led instruction models, game-based learning has a greater chance of successfully facilitating learners to engage in meaningful autonomous learning

[13][14][15][16]. Among different game genres, Alternate Reality Game (ARG) has been widely used in museums, scenic spots, and urban exploration in recent years [17]. This type of game transposes the escape game into a real situation. The interactive plot is often integrated into the real and digital worlds, and the elements of solving puzzles and crimes are incorporated into the story context, making the players feel more immersive and making the game more interesting and topical [18]. University libraries have used escape room games in the past as an application for orientation and outreach programs that can help students become familiar with library services and resources [19][20].

[5] proposed four key elements that a makerspace is regarded as a learning environment, including high playability, high authenticity, development of nuanced knowledge, and linking practice to representation. It is clear that the characteristics of alternative reality games can be applied to makerspace user education, and that augmented reality (AR), a technology that combines virtual and physical environments, can be effective in enhancing students' interest and engagement in learning [21]. [18] suggested that augmented reality games could be used to aid library user education to facilitate students' familiarity with library services; however, no research has been conducted on the use of alternative reality games combined with AR for guided instruction in makerspaces. Therefore, the purpose of this study is to develop a game-based augmented reality navigation system (GARNS) that combines an alternative reality puzzle game and AR tools to support makerspace user education in a university library, allowing learners to understand the use of a makerspace and related equipment through the process of solving puzzles with the support of AR tools, and to compare the difference in learning effectiveness of learners among the use of GARNS, web navigation system, and narrative guided tour with librarian for makerspace user education.

II. RESEARCH METHODOLOGY

A. Research participants

The research participants of this study were 24 11th grade students from a high school in Keelung City, Taiwan. All the research participants were first-time participants in the library's makerspace tour. In this study, the research participants were randomly assigned to the experimental group with GARNS, the control group 1 with a web navigation system, or the control group 2 with a narrative guided tour with librarian for the makerspace user education, including 10 in the experimental group with GARNS, 7 in the control group 1 with the web navigation system, and 7 in the control group 2 with a narrative guided tour with librarian.

B. Experimental design and procedures

In this study, a total of 24 research participants were randomly assigned to either the experimental group supported by GARNS, the control group 1 supported by a web navigation system, or the control group 2 supported by a narrated tour for makerspace user education, and the research participants in all three groups received the same learning content. The experimental group and the control group 1 were offered an individual learning mode for conducting a

makerspace user education through digital learning methods, while the control group 2 was offered a group learning mode for conducting a makerspace user education. The experimental group used the GARNS by a handheld mobile device to learn how to use the equipment of a makerspace to design something through solving game missions and interacting with physical equipment to conduct a makerspace user education, while the control group 1 used a web navigation system by a handheld mobile device and the control group 2 used a narrative guided tour with librarian to learn how to use the equipment of a makerspace to design something to conduct a makerspace user education. The experimental procedures of this study are shown in Figure 1.

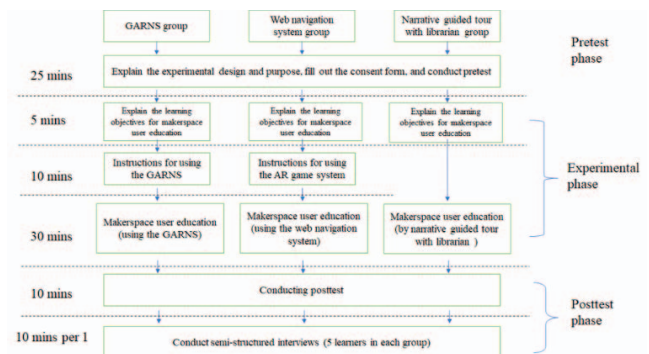


Fig. 1. The experimental procedures of this study

C. Research tools

1) Game-based augmented reality navigation system (GARNS)

This research team used the augmented reality (AR) educational game editing tool “Mini Universe” to design and create a “game-based AR navigation system.” “Mini Universe” was designed with game-based learning and scaffolding theory as its core design framework, providing easy game mechanism setting and content editing functions to help teachers quickly design augmented reality educational games with cognitive scaffolding and real-time feedback mechanisms that are compatible with teaching content to assist learners in exploration and learning. The AR educational game development tool of “Mini Universe” includes two functions: scanning stickers and using stickers: (1) Scanning stickers: “Mini Universe” adopts a broad definition of AR mode. After scanning the object in the reality, additional information can be obtained. Its feature is that the stickers obtained from the AR scanning can be preserved in the game, and users can view the preserved extended supplementary information at any time during the game process through the “use stickers” function. (2) Using stickers: It includes a sticker analysis area and a sticker storage area. The stickers obtained by scanning stickers will be saved and displayed in the sticker storage area. Learners can click on the stickers in the storage area and put them into the analysis area to analyze them with different game mechanisms to view information, solve problems, and complete game tasks.

This study is based on the outline of makerspace user education (as shown in Table 1) to design the learning contents.

Using the makerspace as a learning environment, an augmented reality puzzle-solving game oriented towards problem-solving tasks was developed and evolved into a “game-based AR navigation system” in this study. The game task design concept of this study is to provide learners with the opportunity to explore and solve puzzles in the maker space through the game story. Learners can collect clues by scanning stickers and use them to analyze and complete game tasks, thus learning relevant knowledge about the machine operation. There is total of four game tasks, each corresponding to the learning of a machine. To avoid interference or hitchhiking among multiple players, the game task design has multiple paths. At the beginning of the game, learners will randomly receive the challenge sequence for the four tasks and complete them according to their order. To achieve good learning effectiveness, this study uses clues, matching, and combination as cognitive designs for learning tasks: (1) Clues: learners conduct contextual analysis and inference thinking based on the information they obtain, such as associating the correct items according to the hints given during the game process; (2) Matching: learners can focus on observation and conceptual verification, such as finding the appropriate mold for a specific item; (3) Combination: learners can correctly choose resources and think systematically, such as collecting materials and machines required to make items and combining them to complete game tasks.

TABLE I. THE LEARNING CONTENTS OF THE MAKERSPACE USED IN THIS STUDY

Equipment	Teaching Item
1. FDM 3D printer	1. Machine
2. Stereolithography 3D printer	2. Consumables and appliances
3. Thermal transfer printer	3. File format
4. Laser cutter machine	4. Computer (parameter settings)
	5. Mold

The way that the game missions are carried out is that each learner will receive a “mission guide card” at the beginning, as shown on the front of Figure 2. The top section of the card is the “monster gift recipe,” which prompts learners that each monster’s desired gift is made up of a combination of any type of machine, consumable, file format, computer (software), and mold set. Stickers indicating each category are scattered throughout the makerspace, and learners need to collect the required information and items based on the clues provided. The bottom section is the “monster challenge sequence,” and the challenge order for each learner is different.

The back of the “mission guide card” is shown in Figure 3. When learners have collected all the clues and need to make the desired gift, they must use the blue energy stone at the top to exchange for “energy key” sticker, which allows them to combine the items at the same time and complete the mission. The purpose of the blue energy stone is to prevent learners from simply trying different combinations without actively looking at the content of the stickers to find the answer. Additionally, the blue energy stone can be used to exchange for stickers to obtain additional hints and avoid learners getting stuck in the game, which affects the learning experience. The remaining number of energy stones will be counted in the score, and if they run out, learners will be unable to continue to exchange for “energy keys” and hints.



Fig. 2. A mission guide card- front side

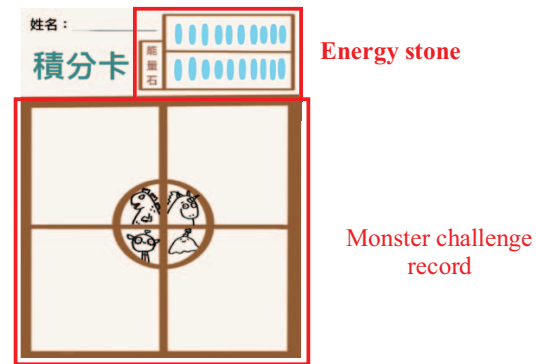


Fig. 3. A mission guide card- opposite side

During the mission completion process, learners use the scanning function to collect stickers (as shown in Figure 4) containing information and tools related to making the gift. Learners can select stickers from the sticker storage area at any time and then click “use alone” (as shown in Figure 5) to view information such as machines, consumables, file formats, and gift clues (as shown in Figure 6). Once learners have collected all the recipe components according to the mission guide, they can use the energy stone to exchange for an energy key sticker and select all the recipe components and energy key stickers to “use at the same time” (as shown in Figure 7). If the combination is correct, a successful combination screen and information will appear (as shown in Figure 8). If the combination is incorrect, learners must recheck or collect the required recipe components and exchange for energy key stickers to try again. After successfully completing the mission, learners can return to the app login screen to continue the challenge with the next monster. Learners who complete the gift challenge of four monsters within the allotted time can pass. Through the puzzle game in

the monster mission, learners can understand the environment, characteristics, and usage of equipment in the makerspace.



Fig. 4. Scanning a sticker



Fig. 5. Using a sticker alone



Fig. 6. Using a sticker to check information



Fig. 7. Simultaneously using stickers



Fig. 8. Successfully using stickers to finish a mission

2) Web navigation system

This study used the Moodle platform to design a web navigation system for the control group 1. The menu of the system interface was designed in two levels according to the learning contents of the makerspace used in this study. The home page will list four learning items including FDM 3D printer tutorial, stereolithography 3D printer tutorial, thermal transfer printer tutorial, and laser cutter machine tutorial (Figure 9), and you can check off the items you have completed. The learning content is presented in the form of a web page with pictures and text descriptions (Figure 10), and learners can select the machine to view its introduction, including machine introduction, usable consumables, file settings, parameter settings, and notes, etc.



Fig. 9. The user menu of web navigation system



Fig. 10. An example of learning content of web navigation system

3) Makerspace user education test sheet

This study compiled the "makerspace user education test sheet" based on the educational content of the designed makerspace, and discussed with the makerspace librarians to determine the appropriateness and validity of the test content. The test content is presented in a problem-solving manner,

providing learners with simulated scenarios and goals to answer questions. The test question types include knowledge-based questions, combination questions, and sorting questions. This study conducted pretests and posttests using this test paper before and after the experimental activities to evaluate learners' prior knowledge and learning effectiveness. The teaching content includes four types of machines, each with five questions. The pretest and posttest will randomly select two types of machines' questions for learners to answer, with a total of 10 questions. The pretest and post-test test papers will randomly select machine types for questions to avoid students answering based on memory.

4) Semi-structure interview

At the end of the experiment, semi-structured interviews were conducted with learners of three groups to further understand their feelings, thoughts, and suggestions on using the GARNs, web navigation system, and narrative guided tour with librarian to conduct makerspace user education in order to supplement the insufficiency of quantitative data analysis.

III. EXPERIMENTAL RESULTS

A. Analysis of difference in learning effectiveness among three groups

Before comparing the differences in learning effectiveness among the three groups, this study first examined whether there was a significant improvement in learning effectiveness within each group. Because the experimental sample size of this study was small, the Wilcoxon sign rank test, a non-parametric statistics method, was used to test whether there was a significant difference in the scores of the "makerspace user education test pretest" and the "makerspace user education test posttest" among the three groups of learners, in order to verify the effectiveness of the learning. The results are shown in Table 2. The test results indicate that the GARNs group ($Z = 2.81$, $p = 0.005 < .01$), web navigation system group ($Z = -2.13$, $p = 0.033 < .05$), and narrative guided tour with librarian group ($Z = -2.38$, $p = 0.018 < .05$) all showed significant differences in pretest and posttest scores within the group, and the posttest scores were all better than the pretest scores. This result shows that learners can achieve significant learning effectiveness through the GARNs, web navigation system, and traditional narrative guided tour with librarian for makerspace user education.

TABLE II. RESULTS OF WILCOXON SIGN RANK TEST OF INTRA-GROUP LEARNING EFFECTIVENESS FOR THE THREE GROUPS

Group	Pretest		Posttest		<i>Z</i>	<i>p</i>
	Average	Standard deviation	Average	Standard deviation		
GARNs group	8.30	3.97	14.60	2.37	2.81**	0.005
Web navigation system group	9.43	2.94	12.00	3.06	-2.13*	0.033
Narrative guided tour with librarian group	8.57	4.35	12.29	2.63	-2.38*	0.018

* $p < .05$, ** $p < .01$

This study further used Kruskal-Wallis test to compare the improvement scores of the three groups of learners before and after learning, in order to examine whether there were significant differences in learning effectiveness among the three groups. Before conducting the Kruskal-Wallis test, Levene's test for homogeneity of variances was first performed. The results of the test did not violate the assumption of homogeneity of variances ($F = 0.42$, $p = 0.660 > .05$), so Kruskal-Wallis test was continued. The analysis results are shown in Table 3. The test results showed that there were significant differences in learning effectiveness among the three groups of learners ($\chi^2 = 6.81$, $p = 0.033 < .05$). Further analysis with Bonferroni post-hoc multiple comparison test showed that there was no significant difference between the GARNs group and the narrative guided tour with librarian group ($Z = 2.59$, $p = 0.212 > .05$), and between the web navigation system group and the narrative guided tour with librarian group ($Z = -1.14$, $p = 1.000 > .05$). This result is different from [22] research, which found that the use of game-based library user education had significantly better learning effectiveness than traditional teaching methods. However, there was a significant difference between the GARNs group and the web navigation system group ($Z = 3.73$, $p = 0.036 < .05$), and the GARNs group was significantly better than the web navigation system group.

TABLE III. RESULTS OF THE KRUSKAL-WALLIS TEST OF INTER-GROUP LEARNING EFFECTIVENESS FOR THE THREE GROUPS

Item	GARNs group (n=10)	Web navigation system group (n=7)	Narrative guided tour with librarian group (n=7)	χ^2	<i>p</i>
	Grade average	Grade average	Grade average		
Progress Score	16.85	8.71	10.07	6.81*	0.033

* $p < .05$

B. Analysis of interview among three groups

From the interview results, it was found that most of the interviewees believed that the learning mode using the GARNs could help them continuously judge, select, and explore learning tasks. The challenges and peer competition inherent in games can help improve learning focus, and the process of exploring and combining stickers to complete learning tasks can effectively help memory, making learners have a deeper impression of the learning content. The interviewees who used the web navigation system for maker space user education stated that learning through mobile devices was very convenient, and the design of the learning content was very detailed and easy to understand. However, there was too much text, and learners may lose patience with the learning process if they cannot be interested in the content, leading to easy forgetting of the learning content. The interviewees who used the narrative guided tour with librarian for makerspace user education stated that the librarian's lead and explanation could allow learners to quickly absorb knowledge. For learners without relevant background knowledge of the makerspace, the librarian's guidance could increase their knowledge of the makerspace and quickly understand the uses of various makerspace machines. The disadvantage is that learning can only be done through the

librarian's explanation and visual observation, and it is often difficult to understand the functions and requirements of each machine in detail due to the limitations of oral explanation.

IV. CONCLUSIONS AND INSTRUCTION SUGGESTIONS

A. Conclusions

This study combines game-based learning with AR tools to develop a GARNS suitable for the characteristics of makerspaces, providing an innovative and effective learning tool. The research results show that learners using the GARNS for makerspace user education achieve significantly better learning effectiveness than those using the web navigation system. Therefore, this study contributes to promoting the popularization of makerspace user education and diverse learning models.

B. Instruction suggestions

Based on the implementation of experiments supporting makerspace user education using the GARNS, this study proposes suggestions for teachers to apply this system in teaching in the future: (1) Adopt a blended learning model, combining three different guide modes: the GARNS, web navigation system, and narrative guided tour with librarian. Use the web navigation system or narrative guided tour with librarian to give learners a preliminary understanding of the makerspace, such as the types of equipment and devices in the space, to help learners use the GARNS more quickly and deepen their learning of makerspace equipment use. At the same time, this approach can address the problem of web-based learners being unable to focus and remember due to too much text, and enhance the detailed understanding of machine equipment for learners guided by staff. (2) The GARNS developed in this study focuses on puzzle-solving game design, and the information presentation interface is mainly based on images provided as clues. If the information presentation method can be changed to 3D objects, it can be used to display the details of machine-made products or the form of collected materials. This can increase learners' impression and interest in learning through 3D objects, and also enhance learners' immersion in the game.

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